Project Proposal

Interpretable and Risk-Aware Autonomous Driving Decisions

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Concept Bottleneck Models with uncertainty aware abstention

Problem and Motivation

Autonomous systems often use neural networks that take sensor data or images and directly produce a decision. That can work well when the input looks similar to the training data, but it creates a problem when the model is wrong: it is hard to tell what part of the reasoning failed. A driving model might choose GO, for example, without giving us a clear way to tell whether it missed a pedestrian, misunderstood the road, or was simply uncertain. For a safety-related system, accuracy by itself is not enough. We also want the system to expose useful information about its decision and to avoid acting confidently when the input is unclear.

Our project will build a small autonomous-driving decision prototype around this idea. Instead of predicting a driving action directly from an image, the model will first predict a few human-readable concepts, such as whether a pedestrian is present, whether a vehicle is present, whether the road is clear, and whether an obstacle appears close. A simple decision model will use those concepts to choose STOP, SLOW, or GO. We will also add an abstention option, meaning the system can output REQUEST HUMAN CONTROL when important concepts are too uncertain. The goal is not to build a complete self-driving system. The goal is to test whether interpretability and uncertainty-aware refusal can make a simplified autonomous decision pipeline easier to inspect and safer when it is unsure.

Related Work

The main idea comes from Concept Bottleneck Models (CBMs). A CBM predicts meaningful intermediate concepts before predicting the final label, which makes it possible to inspect those concepts and even correct them at test time. Their experiments showed that changing incorrect concept predictions can improve the final prediction. Our project uses the same basic idea, but applies it to a simplified road-scene decision problem where the concepts are things a person can understand directly, such as pedestrian presence or road clearance.

The second part of our project is related to selective prediction. Geifman and El-Yaniv (2019) studied models with a reject option, where the model is allowed to avoid making a prediction on examples it does not trust. We use a much simpler version of that idea: if a safety-related concept is too uncertain, the system requests human control rather than forcing STOP, SLOW, or GO. This also connects to the calibration work of Guo et al. (2017), which showed that modern neural networks can be overconfident and that predicted probabilities should not automatically be treated as reliable confidence values. Because of this, we will choose our abstention threshold using validation data rather than assuming that a raw probability is perfectly calibrated.

We also want to see what happens when the input quality gets worse. Hendrycks and Dietterich (2019) showed that common image corruptions such as blur, noise, and weather effects can cause large drops in neural-network performance. We will use a few simple image corruptions for the same reason: a trustworthy system should behave sensibly when the image is less clear than normal. Finally, Rudin (2019) argues that high-stakes systems should use interpretable models when possible instead of relying only on post-hoc explanations of black boxes. That motivates our use of explicit concepts and a shallow decision tree rather than making Grad-CAM or another heatmap the main explanation.

Methodology

We plan to use a small subset of the Cityscapes road-scene dataset, starting with roughly 500 to 1,000 images. Cityscapes includes semantic annotations, so we can derive concept labels without manually labeling every image. The first three concepts will be pedestrian_present, vehicle_present, and cyclist_present. We will also create road_clear and obstacle_close using simple rules based on the lower-center part of the segmentation mask. For example, a large object occupying the lower-center region can be treated as a proxy for a close obstacle. This is not true depth estimation, and we will state that limitation clearly.

A pretrained ResNet18 will be used as the concept predictor. Its final layer will be replaced with five outputs, one for each concept, and the model will be trained as a multi-label classifier. We will first freeze most of the pretrained network and train the new output layer. If needed, we can fine-tune the last ResNet block. The predicted concepts will

then be given to a shallow decision tree with a small maximum depth so that its decision path remains readable. The final driving actions will be STOP, SLOW, and GO.

On top of the CBM, we will add a simple risk rule. If a safety-critical concept falls inside an uncertainty range, the model will abstain and return REQUEST HUMAN CONTROL. The exact threshold will be selected using the validation set. We will also train a second ResNet18 that predicts STOP, SLOW, or GO directly from the image. This will serve as the end-to-end baseline. The important comparison is not only which model has the highest accuracy, but what happens when the models make mistakes and when the input becomes difficult.

A second experiment will test concept intervention. When the CBM makes an incorrect final decision because one of its concept predictions is wrong, we will replace that concept with its ground-truth value and run the decision tree again. For example, if pedestrian_present is predicted as false but the annotation says true, we can correct that one concept and check whether the action changes from GO or SLOW to STOP. This gives us a direct way to measure whether the interpretability built into the model is actually useful rather than only producing an explanation after the fact.

Evaluation Plan

We will evaluate the project on a held-out test set and keep threshold selection and other tuning on the validation set. For the concept model, we will report precision, recall, and F1 score for each of the five concepts. For the final decision task, we will report accuracy and macro F1 for the standard CBM and the end-to-end baseline.

The main trustworthy-AI metrics will focus on failure behavior. Intervention Success Rate will measure the percentage of incorrect actions that become correct after an incorrect concept is replaced with its ground-truth value. For the risk-aware model, Coverage will measure the percentage of test examples on which the model actually makes a driving decision instead of abstaining. We will also report the error rate on the non-abstained examples and count severe errors where the ground-truth action is STOP but the model chooses GO without abstaining. This lets us see the trade-off between making fewer decisions and making safer decisions.

Finally, we will create corrupted versions of part of the test set using a small number of common changes such as blur, darkness, and image noise. We will compare how much the action accuracy drops and whether the risk-aware model abstains more often on those harder inputs. A useful result would be that the risk-aware model gives up some coverage but reduces the number of severe confident errors.

Course Themes

This project combines at least two major trustworthy-AI themes. The first is interpretability and explainability: the Concept Bottleneck Model exposes intermediate concepts and the shallow decision tree gives a readable path from those concepts to the action. The second is uncertainty, robustness, and safe failure: the abstention mechanism allows the system to hand control back when the visual evidence is unclear, and the corruption tests examine how behavior changes under degraded inputs. Human oversight is also part of the project because concept intervention allows a person to correct an intermediate prediction and observe the effect on the final decision.

The project is intentionally small enough for a semester. We are not implementing a simulator, real steering or throttle control, LiDAR, reinforcement learning, or a new object detector. If the full plan becomes too large, the minimum version will keep the CBM, the end-to-end baseline, one abstention rule, and the concept-intervention experiment.

References

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